

The Solvability Frontier: Which Open Problems Sit Next to Solved Ones, and a Backtest of the Rule

Bucket Foundation

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Abstract

We embed 5,080 problem statements across seven canon branches with a small sentence encoder, call a problem’s **reach** its highest similarity to a solved problem, and draw a **frontier** at the 10th percentile of solved problems’ reach, 0.781 on this set [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. Open problems inside the frontier are sorted into a **reach class**: 251 problems where known results sit close, 764 borderline, 911 that need a new idea and 103 in a branch with too few solved rows to judge [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. A **backtest** at cutoffs 2005 and 2021 scores the rule on the questions posed by those years: at 2005 the inside rate of resolution by 2026 is 46% against 61% outside (ratio 0.752, permutation $p = 0.051$, AUC 0.479, $n = 183$); at 2021 it is 43% against 39% (ratio 1.102, $p = 0.503$, AUC 0.523, $n = 364$) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The rule carries no measurable signal about resolution on this data, and the paper says why: similarity measures vocabulary, resolved years exist for 224 of 1,621 solved rows, the tested rows come from two cohorts, and 50 stored neighbours bound the reach at a cutoff.

1 Introduction

Bucket’s solvability atlas lists open and closed problems with their statements, branch, status and, where known, the years they were posed and resolved. The question behind the frontier is whether an open problem that sits close in meaning to solved problems is closer to being solved, and whether an AI system working from known results could reach it. The frontier drawing answers the first clause by a rule; this paper tests the rule on problems whose outcome is known and publishes the ranked predictions for the open ones, with every number tagged to the file and command that reproduces it.

2 Related work

The atlas draws on the 2005 Science list of 125 questions [1, 2], the 2021 Science and Shanghai Jiao Tong University list¹, the google-deepmind formal-conjectures repository² and the Wikipedia lists of unsolved problems³. The encoder is bge-small-en-v1.5, whose model card points at the C-Pack paper [3]. Systems that generate and test scientific hypotheses end to end [4, 5] are the setting in which a reach score would be used; this paper measures only whether the score tracks outcomes.

¹<https://www.science.org/content/resource/125-questions-exploration-and-discovery>, read from a Wayback copy of the booklet; no DOI.

²<https://github.com/google-deepmind/formal-conjectures>, Apache-2.0; no DOI.

³https://en.wikipedia.org/wiki/Lists_of_unsolved_problems, CC BY-SA 4.0; no DOI.

3 Preliminaries

Definition 3.1 (Set). The set is the 71 atlas problems and the 5,009 rows of `problems-sourced.tsv`, 5,080 rows once Lean theorems are left off [empirical: `output/solvability-frontier/report/frontier.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. A top-level row is solved when its status is solved. A variant is solved only when it and its parent are both solved. Each row’s statement is embedded with `bge-small-en-v1.5` at revision `5c38ec7`, and its 50 most similar rows are stored with cosine similarity to three decimals [empirical: `output/solvability-frontier/report/frontier.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Definition 3.2 (Reach). The `reach` of a row is its highest similarity to a solved row other than itself, taken over its stored neighbours and its stored nearest solved row.

Definition 3.3 (Frontier). The `frontier` threshold τ is the 10th percentile of reach over the solved rows. A row with reach at or above τ is inside; below it, outside. An open row whose branch holds fewer than 10 solved rows is unsampled and counted outside.

Definition 3.4 (Growth). For a top-level row beyond the frontier, `growth` is one plus the number of beyond top-level rows among its 50 stored neighbours whose similarity to it is at or above τ : the number of rows that would move inside if it were solved. Variants neither pull nor count as pulled.

Definition 3.5 (Reach class). Let u be the 75th percentile of reach among open rows inside the frontier. An open top-level row is “AI can reach with known results” when inside with reach at or above u , “borderline” when inside below u , “needs a new idea” when beyond, and “unsampled” when its branch holds fewer than 10 solved rows. Within a class, rows rank by growth, then reach, then id; a ranking by score with unique ids has one ordering [bm:BucketMath.Ranking.rank_ordered].

Definition 3.6 (Backtest). For a cutoff year c , the solved set S_c holds every row solved under 3.1 with a resolved year at or before c . The tested set holds every row posed at or before c and outside S_c . Reach is computed as in 3.2 against S_c alone. When none of a row’s stored neighbours is in S_c , its reach is unknown and bounded above by its 50th neighbour’s similarity. The threshold τ_c is the 10th percentile of reach over the rows of S_c that have a neighbour in S_c ; the rest are counted and dropped. A tested row is inside when its reach is at or above τ_c , outside when its reach or its bound is below τ_c , and undecided otherwise; undecided rows are counted and left out of the rates. The outcome is the row’s 2026 status: solved or partial counts as resolved, open does not.

4 Method

Statistics. Per cutoff and per stratum we report the inside and outside counts, the resolved rate on each side, their ratio (inside over outside, undefined when the outside rate is zero [bm:BucketMath.Marketing.ratio]) and a two-sided permutation p -value: the outcomes are shuffled 10,000 times under seed 20261006 and p is the share of shuffles whose absolute rate difference is at least the observed one, with one added to numerator and denominator [empirical: `output/solvability-frontier/report/backtest.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The p -value so defined is positive and at most one (C). A stratum with fewer than 10 rows on either side reports counts only. The AUC is the Mann-Whitney probability that a resolved row’s reach exceeds an unresolved row’s, ties counting one half.

Strata. Rows are stratified by the word count of the embedded text, under 20, 20 to 40 and over 40 words, and by branch. The bands partition the naturals (C). The length bands exist because the dated lists are short paraphrases: their statements average 19 words against 34 for the rest of the file [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Reproduction. tools/solvability-atlas/build.sh rebuilds the embeddings and the neighbour file; npx ts-node scripts/build-solvability-frontier-report.ts writes frontier.json, backtest.json, predictions.json and predictions.csv under output/solvability-frontier/report/ and the page data under src/lib/research-os/. make figures in this directory redraws every figure and table from those files.

5 Results

5.1 The frontier

The frontier sits at $\tau = 0.781$: 1,621 rows solved, 2,225 reachable, 1,131 beyond and 103 unsampled, 3,846 inside and 1,234 outside [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. 1 gives the split per branch. Mathematics holds 4,112 of the 5,080 rows and 1,473 of the 1,621 solved ones [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts], so it sets the threshold on its own.

Table 1: Rows per branch and zone [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Branch	Total	Solved	Reachable	Beyond	Unsampled	Inside	Outside
mathematics	4112	1473	2171	468	0	3644	468
physics	259	37	28	194	0	65	194
biophysics	208	21	4	183	0	25	183
cosmology	127	16	0	111	0	16	111
information	115	29	19	67	0	48	67
mind	108	5	0	0	103	5	103
chemistry	85	13	3	69	0	16	69
applied	66	27	0	39	0	27	39

5.2 Backtest

2, 3 and 2 hold the backtest. At cutoff 2005, 108 rows were solved by then, 16 of them with no solved neighbour among the 50 stored and dropped from the threshold, which came out at 0.644; 287 rows were posed by 2005 and open at the cutoff, 38 unsampled, 66 undecided and 183 scored [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The inside rate is 46% over 112 rows against 61% over 71 outside, ratio 0.752, $p = 0.051$, AUC 0.479 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. At cutoff 2021, 190 rows were solved by then, 11 dropped, threshold 0.656; 465 rows tested, 58 unsampled, 43 undecided and 364 scored [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The inside rate is 43% over 238 against



Figure 1: The frontier on 5,080 rows. Solved rows fill the inner disc by reach, open rows inside fill the ring up to the circle at 0.781, outside rows sit beyond it by how far their reach falls short, and unsampled rows are grey crosses [empirical: `output/solvability-frontier/report/frontier.json`, 2026-10-06, `npix ts-node scripts/build-solvability-frontier-report.ts`]. Angle is the row’s rank along the first two principal components of the embeddings.

39% over 126, ratio 1.102, $p = 0.503$, AUC 0.523 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Table 2: Backtest at cutoff 2005, by stratum [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	112	71	46%	61%	0.752	0.051
under 20 words	28	20	46%	70%	0.663	0.139
20 to 40 words	44	36	46%	75%	0.606	0.013
over 40 words	40	15	45%	13%	3.375	0.053
applied	4	3	under 10 on one side			
biophysics	18	44	78%	73%	1.069	0.756
chemistry	7	2	under 10 on one side			
cosmology	11	5	under 10 on one side			
mathematics	49	1	under 10 on one side			
physics	23	16	39%	31%	1.252	0.745

Table 3: Backtest at cutoff 2021, by stratum [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	238	126	43%	39%	1.102	0.503
under 20 words	101	57	37%	37%	0.994	1.000
20 to 40 words	77	49	49%	57%	0.864	0.465
over 40 words	60	20	45%	0%		0.000
applied	13	11	15%	18%	0.846	1.000
biophysics	39	52	64%	67%	0.952	0.823
chemistry	30	16	20%	19%	1.067	1.000
cosmology	20	10	40%	40%	1.000	1.000
information	18	11	33%	9%	3.667	0.197
mathematics	87	1	under 10 on one side			
physics	31	25	32%	16%	2.016	0.223

Two strata reach p below 0.05, and both run against each other: at 2005 the 20 to 40 word band resolves at 46% inside against 75% outside ($p = 0.013$), while at 2021 the over 40 word band resolves at 45% inside against 0% outside over 20 rows ($p = 0.000$) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The second is a cohort effect: the long statements outside at 2021 are the 2021 list’s own longer paraphrases, all still open five years on. Mathematics scores one row outside at both cutoffs, because nearly every dated mathematics row has a solved mathematics neighbour, so no mathematics rate is given.

5.3 Predictions

4 gives the class counts: 251 rows where AI can reach with known results, 764 borderline, 911 that need a new idea and 103 unsampled, with $u = 0.854$ [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. 5, 6, 7 and 8 list the first ten of each class by growth; the full ranked table with the three nearest solved problems and the starting works is predictions.csv. B lists the 56 open atlas problems in full. Among them, Goldbach and

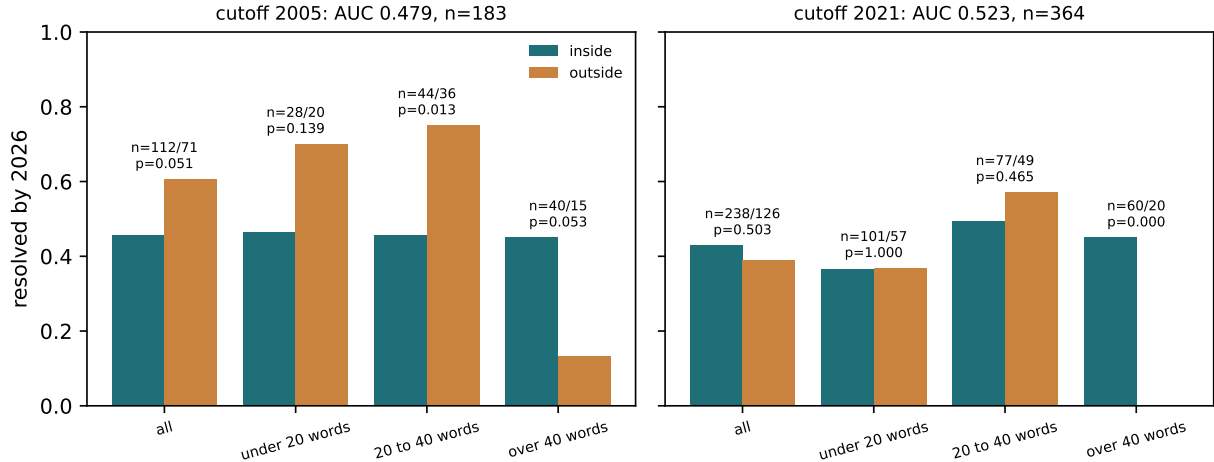


Figure 2: Resolved rate by 2026 inside and outside the cutoff frontier, over all scored rows and by length band [empirical: `output/solvability-frontier/report/backtest.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

the Langlands program fall in the first class, with reach 0.867 and 0.863 to an Erdős problem and to the geometric Langlands conjecture [empirical: `output/solvability-frontier/report/predictions.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]; the Riemann hypothesis, Navier-Stokes, dark matter, weak cosmic censorship and quantum gravity head the “needs a new idea” class by growth, 4, 4, 3, 3 and 3 [empirical: `output/solvability-frontier/report/predictions.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The Goldbach placement is the vocabulary effect of 7: the nearest solved row shares the words, and nothing in the method says whether it shares a method.

Table 4: Open top-level problems per reach class [empirical: `output/solvability-frontier/report/predictions.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Class	Open problems
AI can reach with known results	251
borderline	764
needs a new idea	911
unsampled	103

6 Calibration

The central claims are empirical, so the calibration is the backtest itself: the rule is fixed before the outcomes are read, the outcomes are the 2026 status column with its per-row source, and the threshold and the strata are recomputed at each cutoff from data dated at or before it. The error budget is the permutation p -value and the stratum floor. The two cutoffs agree on the sign of nothing: the ratio is 0.752 at 2005 and 1.102 at 2021, and the AUC sits at 0.479 and 0.523, each within 0.03 of chance [empirical: `output/solvability-frontier/report/backtest.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The reading is that reach, as defined here, does not

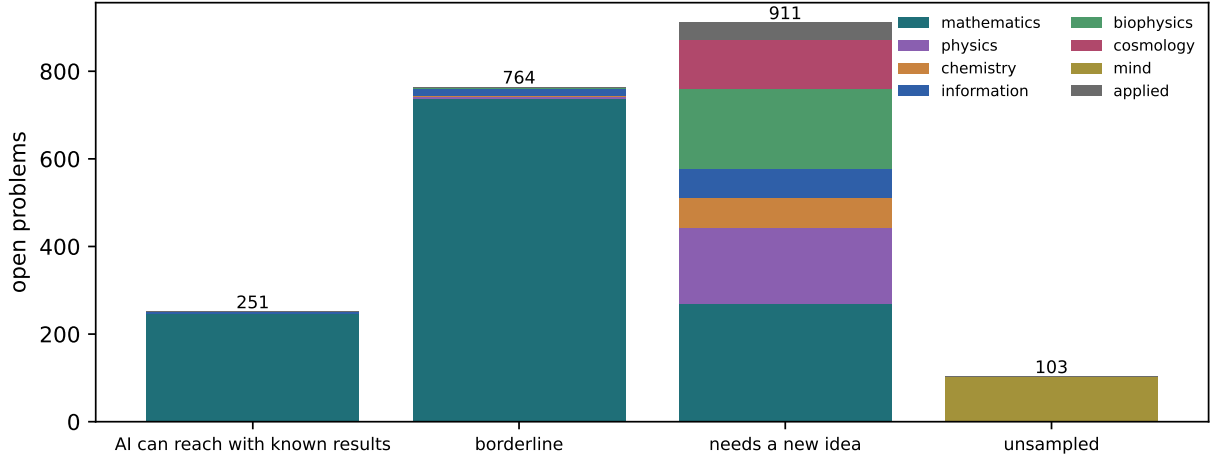


Figure 3: Reach classes by branch [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, `npm run ts-node scripts/build-solvability-frontier-report.ts`].

Table 5: AI can reach with known results, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, `npm run ts-node scripts/build-solvability-frontier-report.ts`].

Problem	Branch	Reach	Growth	Nearest solved
Pierpont Prime, infinitely many pierpont primes	mathematics	0.870	4	Regular Primes, infinitude of irregularprimes
Can the graph isomorphism problem be solved in polynomial time on a classical computer?	information	0.924	3	Graph isomorphism in quasi-polynomial time
Wieferich Prime, infinite isWieferichPrime	mathematics	0.871	3	Regular Primes, infinitude of irregularprimes
OEIS A945, every prime occurs	mathematics	0.868	2	OEIS A363102, a eq one or prime
Factorial Prime, infinitely many factorial primes	mathematics	0.908	1	Regular Primes, infinitude of irregularprimes
Caratheodory conjecture	mathematics	0.894	1	Caratheodory conjecture for surfaces of smoothness C^4 (Brendan Guil)
Is graph canonization polynomial-time equivalent to the graph isomorphism problem?	information	0.887	1	Graph isomorphism in quasi-polynomial time
Erdos problem 1030	mathematics	0.882	1	Erdos problem 1014
Erdos problem 968	mathematics	0.882	1	Erdos problem 822
OEIS A93818	mathematics	0.878	1	OEIS A372761, exists unique a eq prime

Table 6: Borderline, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Problem	Branch	Reach	Growth	Nearest solved
The density hypothesis for zeroes of the Riemann zeta function	mathematics	0.795	6	Erdos problem 12, erdos sarkozy density 0
Grand Riemann hypothesis	mathematics	0.789	6	Poisson Conjecture
Are there infinitely many Pell primes?	mathematics	0.804	5	Regular Primes, infinitude of irregularprimes
Can integer factorization be done in polynomial time?	mathematics	0.804	5	Linear programming in polynomial time
Are there infinitely many Wolstenholme primes?	mathematics	0.796	5	Regular Primes, infinitude of irregularprimes
Are there infinitely many Kynea primes?	mathematics	0.782	5	Regular Primes, infinitude of irregularprimes
The covering problem of Rado	mathematics	0.824	4	Erdos problem 1071, i
Hilbert?Polya conjecture	mathematics	0.820	4	OEIS A1818, conjecture1
Woodin's Ω -conjecture	mathematics	0.811	4	Erdos problem 1036, alon hajnal
Are there infinitely many Wieferich primes?	mathematics	0.809	4	OEIS A159829, conjecture1

Table 7: Needs a new idea, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Problem	Branch	Reach	Growth	Nearest solved
Cosmological principle	physics	0.721	8	Baryon acoustic oscillations
Extended Riemann Hypothesis	mathematics	0.778	6	Riemann Zeta Values, infinite irrational at odd
Generalized Riemann hypothesis for Selberg class	mathematics	0.777	6	Erdos problem 193, z2
L = P problem	information	0.758	6	Erdos problem 318, univ
Dark energy	physics	0.663	6	Photon underproduction crisis (2014?2015)
Riemann Hypothesis, riemannHypothesis	mathematics	0.776	5	OEIS A49473
Generalized Riemann hypothesis	mathematics	0.773	5	Erdos problem 990, hayman
Are there infinitely many sexy primes?	mathematics	0.769	5	Regular Primes, infinitude of irregularprimes
Supermassive black holes	physics	0.757	5	Black hole at the Galactic center
Black holes, black hole information paradox, and black hole radiation	physics	0.740	5	Ultraviolet catastrophe

Table 8: Unsourced, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Problem	Branch	Reach	Growth	Nearest solved
What is reflected in sensory neocortical activity?	mind	0.724	0	Neurons that encode location
Where do our memories get stored and how are they retrieved again?	mind	0.713	0	Cellular basis of memory storage
Is there a canonical computation performed by cortical columns?	mind	0.701	0	Erdos problem 298, natural density
Can human memory be stored, manipulated, and transplanted digitally?	mind	0.700	0	Cellular basis of memory storage
Connectionism	mind	0.699	0	Keller’s conjecture (Joshua Brakensiek, Marijn Heule, John Mackey, Dav
Modularity of mind	mind	0.699	0	Cellular basis of memory storage
How Are Memories Stored and Retrieved?	mind	0.697	0	Cellular basis of memory storage
Free will, particularly the neuroscience of free will	mind	0.696	0	Keller’s conjecture (Joshua Brakensiek, Marijn Heule, John Mackey, Dav
Can we understand the action of brain in natural environments?	mind	0.695	0	Chemical neurotransmission at synapses
How do neurons compute?	mind	0.692	0	Chemical neurotransmission at synapses

predict which open problem gets resolved on this data.

7 Limitations

Vocabulary. Cosine similarity between statement embeddings measures shared words and phrasing. Two problems can share a method and no vocabulary, or vocabulary and no method.

Mathematics. 4,112 of 5,080 rows and 1,473 of 1,621 solved rows are mathematics [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The threshold is a mathematics quantity, and the other six branches are read against it.

Two cohorts. Of the 517 rows with a posed year, 241 come from the 2005 and 2021 Science lists, and those two lists supply most of the tested rows at both cutoffs [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. A list’s paraphrases share length and style, which is what the length bands are for, and the over 40 word result at 2021 shows the cohort rather than the rule.

Partial as resolved. Many list questions saw an advance without a settlement, and the outcome calls both resolved. A solved-only outcome leaves too few resolved rows for a rate.

The 50-neighbour bound. At a cutoff with few solved rows, most of a row’s solved peers fall outside its 50 stored neighbours, so its reach is unknown: 66 rows undecided and 16 solved rows dropped at 2005, 43 and 11 at 2021 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. Dropping them can only raise the threshold.

Resolved years. Only 224 of the 1,621 solved rows carry a resolved year, so the solved set at a cut-off, 108 at 2005 and 190 at 2021, is a fraction of what was solved by then [empirical: `output/solvability-frontier/report/backtest.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Sources and licences. `problems.tsv` is MIT. formal-conjectures rows are Apache-2.0. Wikipedia lists and the DARPA and XPRIZE rows are CC BY-SA 4.0 with attribution in the data. The Science 2005 and 2021 lists, the systems-neuroscience chapter titles and the Holy Grails titles are cited as headlines under 15 words with paraphrased statements, since the bodies are under copyright. The encoder is MIT.

References

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- [5] Chris Lu et al. “The AI Scientist: towards fully automated open-ended scientific discovery”. In: (2024). arXiv: [2408.06292](https://arxiv.org/abs/2408.06292).

A Glossary

reach <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-frontier.ts>

A problem’s highest cosine similarity to a solved problem other than itself, over its stored neighbours; see 3.2.

frontier <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-frontier.ts>

The reach value at the 10th percentile of solved problems’ reach; a problem at or above it is inside, below it outside; see 3.3.

growth <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-frontier.ts>

For a problem beyond the frontier, one plus the number of beyond problems among its stored neighbours that sit at or above the threshold from it; see 3.4.

reach class <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-predictions.ts>

One of four labels for an open problem: AI can reach with known results, borderline, needs a new idea, unsampled; see 3.5.

backtest <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-backtest.ts>

The frontier rule applied at a past cutoff year to the problems posed by then, scored against their 2026 status; see 3.6.

B The atlas problems

Problem	Class	Branch	Reach	Growth	Nearest solved
Goldbach conjecture	reach	mathematics	0.867	0	Erdos problem 418, conditional
Langlands program	reach	mathematics	0.863	0	Geometric Langlands conjecture
Graph isomorphism complexity	borderline	information	0.833	0	Graph isomorphism in quasi-polynomial time
Jacobian conjecture	borderline	mathematics	0.807	0	Poisson Conjecture, jacobian implication
Collatz conjecture	borderline	mathematics	0.786	0	OEIS A306424, forty three is max
Riemann hypothesis	needs a new idea	mathematics	0.771	4	Deterministic polynomial-time primality testing
Navier-Stokes existence and smoothness	needs a new idea	physics	0.671	4	Poincare, smooth known cases
Dark matter identity	needs a new idea	cosmology	0.678	3	Baryon acoustic oscillations
Weak cosmic censorship	needs a new idea	cosmology	0.659	3	Baryon acoustic oscillations
Quantum gravity	needs a new idea	cosmology	0.615	3	Microscopic Weighting, iff finite concentration
Room-temperature superconductivity mechanism	needs a new idea	physics	0.603	3	Element predicted as eka-aluminium
Programmed versus stochastic aging	needs a new idea	biophysics	0.602	3	Carrier of hereditary information
Glass transition	needs a new idea	chemistry	0.601	3	Continuum hypothesis independence
Integer factoring complexity	needs a new idea	information	0.752	2	OEIS A382590, kthPrimeFactor periodic
Hadwiger-Nelson problem	needs a new idea	mathematics	0.743	2	Erdos problem 780
Invariant subspace problem	needs a new idea	mathematics	0.727	2	Green open problem 49
Matrix multiplication exponent omega	needs a new idea	information	0.725	2	OEIS A108306
Existence of one-way functions	needs a new idea	information	0.711	2	Erdos problem 198, concrete
Hubble tension	needs a new idea	cosmology	0.707	2	Expansion of the universe
BQP vs BPP separation	needs a new idea	information	0.693	2	Graph isomorphism in quasi-polynomial time
Yang-Mills mass gap	needs a new idea	physics	0.693	2	Microscopic Weighting, iff finite concentration
Black hole information paradox	needs a new idea	cosmology	0.691	2	Ultraviolet catastrophe
Origin of life	needs a new idea	biophysics	0.688	2	Urea synthesis from inorganic reagents
Cosmological constant problem	needs a new idea	cosmology	0.668	2	Erdos problem 1005, constant
Quantum measurement problem	needs a new idea	physics	0.637	2	Einstein problem (David Smith, Joseph Samuel Myers, Craig S
Birch and Swinnerton-Dyer conjecture	needs a new idea	mathematics	0.764	1	Elliptic Curve Rank, goldfeld two le
abc conjecture	needs a new idea	mathematics	0.760	1	Erdos problem 494, k eq 3 card gt 6
Unique Games Conjecture	needs a new idea	information	0.751	1	Complexity of computing Nash equilibria
Twin prime conjecture	needs a new idea	mathematics	0.745	1	OEIS A91669, prime and primitive root of dvd

Problem	Class	Branch	Reach	Growth	Nearest solved
P vs NP	needs a new idea	information	0.732	1	Complexity of computing Nash equilibria
Protein folding pathway kinetics	needs a new idea	biophysics	0.721	1	Protein structure prediction
Hodge conjecture	needs a new idea	mathematics	0.714	1	Poincare conjecture
Hadamard matrix conjecture	needs a new idea	mathematics	0.706	1	Gerstenhaber Three Matrices, exists finrank adjoin quadruple
Succinct proofs without trusted setup optimality	needs a new idea	information	0.702	1	Erdos problem 967, tauberian
Elliptic curve discrete log hardness	needs a new idea	information	0.697	1	Modularity Conjecture
Exact exchange-correlation functional	needs a new idea	chemistry	0.683	1	Erdos problem 298, natural density
Bioelectric pattern control	needs a new idea	biophysics	0.681	1	Chemical neurotransmission at synapses
Market efficiency as computational hardness	needs a new idea	information	0.674	1	Erdos problem 298, natural density
Fermion sign problem	needs a new idea	physics	0.666	1	Monochromatic Quantum Graph, eqSystem6 no solution d5
Anomalous properties of water	needs a new idea	chemistry	0.659	1	Proof of the atomic theory
Climate sensitivity bound	needs a new idea	physics	0.653	1	Sensitivity conjecture
Autoformalization of mathematics	needs a new idea	information	0.649	1	21 149, kourouka 21 149 not inner
Optimal multi-item auction design	needs a new idea	information	0.649	1	Complexity of computing Nash equilibria
2D Hubbard model ground state	needs a new idea	physics	0.640	1	Existence of time crystals (2012?2016)
Formal reasoning limits of transformers	needs a new idea	information	0.637	1	Erdos problem 152
Mitochondrial redox and aging mechanism	needs a new idea	biophysics	0.634	1	Carrier of hereditary information
AI alignment verification	needs a new idea	information	0.624	1	Pandemic Response XPRIZE
Rough volatility pricing	needs a new idea	mathematics	0.622	1	Pricing of options
Turbulence closure	needs a new idea	physics	0.617	1	Voronovskaja Type Formula, bezier bernstein operators
Rational catalyst design	needs a new idea	chemistry	0.598	1	Complexity of computing Nash equilibria
Plasma confinement stability	needs a new idea	physics	0.588	1	Solar abundance problem (1998?2022)
Light, circadian biology and metabolism	needs a new idea	biophysics	0.572	1	OEIS A100478
Smale's 18th problem limits of intelligence	unsampled	mind	0.673	0	Hilbert's tenth problem
Neural binding problem	unsampled	mind	0.660	0	Controlling neurons with light
Free energy principle formalization	unsampled	mind	0.659	0	Erdos problem 298, natural density
Hard problem of consciousness	unsampled	mind	0.641	0	Keller's conjecture (Joshua Brakensiek, Marijn Heule, John M

C Lean listings

`lean/Bucket/Backtest.lean`, checked by `make lean`, holds the two facts the method leans on: the corrected permutation p has a positive numerator no larger than its denominator, and the length bands cover every word count.

```

structure PermutationP where
  hits : Nat
  permutations : Nat

```

```

bounded : hits <= permutations

def PermutationP.numerator (p : PermutationP) : Nat := p.hits + 1
def PermutationP.denominator (p : PermutationP) : Nat := p.permutations
+ 1

theorem PermutationP.numerator_pos (p : PermutationP) : 0 < p.numerator
:=
Nat.succ_pos p.hits

theorem PermutationP.numerator_le_denominator (p : PermutationP) :
p.numerator <= p.denominator :=
Nat.succ_le_succ p.bounded

def lengthBand (words : Nat) : LengthBand :=
if words < 20 then .short else if words <= 40 then .middle else .long

theorem lengthBand_total (words : Nat) :
lengthBand words = .short \ / lengthBand words = .middle \ /
lengthBand words = .long

```

lake build output, quoted verbatim.

Build completed successfully (4 jobs).